Dirt Bike Maintenance Sensor System

Final Presentation

CSC494

The Problem

Dirt bikes face extreme conditions:

- Dust & mud
- High RPMs
- Crashes & tipping

80-90% of engine failures are caused by neglected maintenance.

Main causes:

- Clogged air filter
- Low oil
- Overheating

Riders usually don’t notice until damage is already done.

Problem Domain — In My Own Words

I ride a 2018 Yamaha YZ450F. Checking the air filter or oil temperature mid-ride just doesn’t happen — you’re focused on riding.

The air filter clogs with mud and dust, and the oil gets too hot, but there’s no warning until the engine loses power or seizes.

The problem I solved:
Give riders real-time alerts on their phone so they can prevent expensive damage before it happens.

Engine rebuilds cost **$1,000-$3,000**. This system helps avoid that.

The Solution

A real-time IoT sensor system that monitors:

- **Air filter restriction** using a differential pressure sensor
- **Oil temperature** using a waterproof probe

Key Features:

- Zero permanent modifications to the bike
- Bluetooth Low Energy data sent to phone
- Live gauges and threshold alerts on a web dashboard
- Powered from the bike's existing SAE trickle charger plug

Solution Domain — In My Own Words

I built a sensor system that mounts without drilling or permanent changes.

- Pressure sensor tube routes through the air box lid seam
- Temperature probe clamps to the engine case near the oil fill cap

Data is processed by a small microcontroller and broadcast via Bluetooth.
A web dashboard (no app install) shows live gauges, color-coded alerts, and history charts.
It opens in Bluefy on iPhone.

The entire system plugs into the bike’s SAE connector — truly plug-and-play.

Original Plan vs Reality

What I Planned	What Happened
Ultrasonic oil level sensor	Dropped — too invasive
Coolant level sensor	Dropped — no non-invasive option
WiFi cloud sync	Not needed — BLE was enough
DC-DC converter from 12V	Replaced with SAE to USB adapter
Pressure sensor + dashboard	✅ Fully implemented
Push alerts	✅ Implemented via BLE

Lesson: Keep it simple and non-invasive for the prototype.

Technology Stack

- **XIAO ESP32-C6** — Main MCU + Bluetooth 5.0
- **Arduino Nano** — Reads DS18B20 1-Wire protocol
- **MPXV7002DP** — Air filter differential pressure
- **DS18B20** — Waterproof oil temperature probe
- **Standard ESP32 BLE** — Better iOS compatibility
- **ArduinoJson** — Lightweight JSON telemetry
- **HTML + CSS + JS + Web Bluetooth** — Mobile dashboard
- **GitHub Pages** — Free HTTPS hosting
- **SAE to USB adapter** — Bike power source

How AI Helped

AI was very helpful for:

- Fast debugging using serial monitor output
- Generating exact wiring tables with pin numbers
- Rewriting specific firmware functions
- Quickly identifying hardware issues

Where AI was wrong:

- Suggested `analogSetAttenuation()` (doesn't exist on XIAO ESP32-C6)
- Outdated NimBLE callback signatures
- Didn't know ESP32-C6 RISC-V core can't bit-bang 1-Wire

Lesson: AI is a powerful tool, but always test and verify.

Learning with AI - Topic 1: Sensor Interfacing

What I set out to learn:

How to connect, calibrate, and mount sensors on a real dirt bike.

What I learned:

- Real hardware is very different from tutorials
- Pressure sensor failed on A0 → moved to A1
- ESP32-C6 RISC-V cannot reliably bit-bang 1-Wire for DS18B20 → offloaded to Arduino Nano
- Mounting on a vibrating dirt bike (zip ties, foil, air box seam) is much harder than breadboard work

Real-world vibration, dust, and “no drilling” rule made everything more challenging.

Learning with AI - Topic 2: The Dashboard

What I set out to learn:

How to build a wireless mobile dashboard with live data and alerts.

What I learned:

- Web Bluetooth does **not** work in Safari on iPhone
- NimBLE library was invisible to iOS → switched to standard ESP32 BLE library
- Chrome on Windows blocked it (no Bluetooth adapter)
- Final solution: GitHub Pages (HTTPS) + Bluefy browser on iPhone

Browser security, platform limits, and hosting requirements made it far more complex than expected.

Weekly Progress Summary

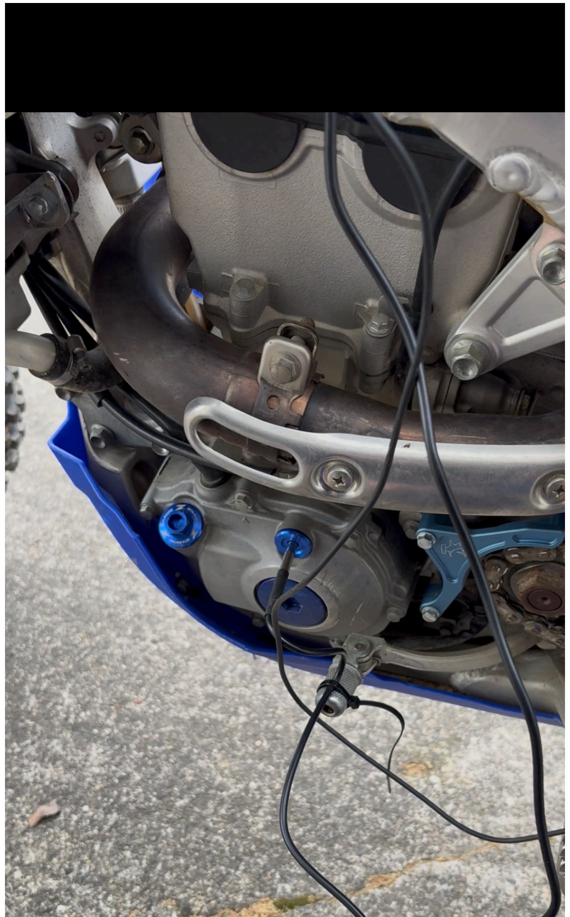
Stage 1 (Weeks 1-3): Research, requirements, and GitHub setup

Sprint 1 (Weeks 4-7): Hardware, firmware, BLE, and first presentation

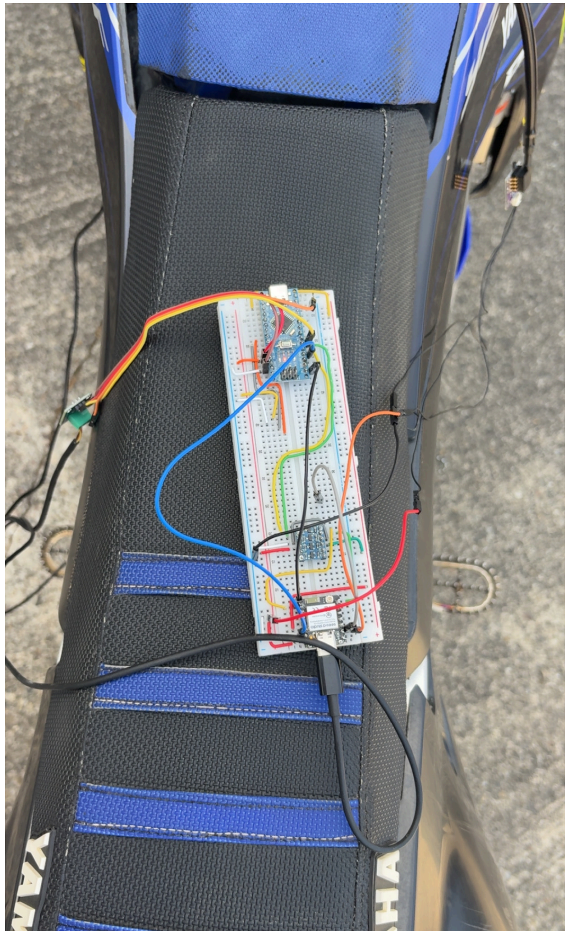
Sprint 2 (Weeks 10-15): Dashboard development, bike testing, and final deliverables

Hardware Setup

Non-invasive installation on the 2018 Yamaha YZ450F

Pressure Sensor	Oil Temp Probe
	

Electronics & Power

Breadboard	SAE Power
	

Demonstration

Live Demo Videos

Cold Start

▶️ Watch Cold Start

Warm Engine

▶️ Watch Warm Engine

What the videos show:

- Real-time dashboard connected via Bluetooth in Bluefy
- Pressure sensor reacting to throttle
- Oil temperature increasing during warm-up
- Threshold alerts activating correctly

Final Results

System works successfully on the 2018 Yamaha YZ450F!

- ✅ Pressure sensor responds to throttle and restriction
- ✅ Temperature sensor tracks warm-up from ~20°C
- ✅ Bluetooth device visible as `DirtBike_001`
- ✅ Live web dashboard with alerts in Bluefy
- ✅ Powered directly from bike's SAE connector
- ✅ All code and docs on GitHub

Future Improvements

- Improved mounting (hose clamps, thermal paste, barbed fittings)
- Weatherproof enclosure
- Solder or heat shrink tubing for better connection
- Custom oil fill cap for direct oil temperature
- Machine learning for predictive alerts
- Support for bikes without batteries